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NAAC GRADE A
Department of Computer Science & Engineering

CERTIFICATE

Advanced UI/UX | V Semester B.E.

This is to certify that

Joseph Stalin D

bearing USN [1EP24CS053], Section A, has successfully completed the Advanced UI/UX activity (Problem No. 24) titled

"Collaborative Design Tool Conflict Handling"

& has submitted this report as required on [11-11-2026], during the academic year 2026-27.

Prasanth
Assistant Professor,
Dept. of CSE

Date of Submission:

Abstract

Real-time collaborative design tools let many people edit one canvas, but they handle simultaneous edits on the same object poorly. Users lose work silently, cannot tell who is editing what, and fear pressing undo because it may erase a teammate's changes. This project designs how such a tool shows and resolves conflicting edits. Research with [N] designers and students through interviews and observation, plus analysis of Figma, Canva and Miro, showed that users want early awareness of others' activity, clear visibility when a conflict occurs, and predictable undo. Two personas and an empathy map led to the design question: how might we let people edit together without fear of overwriting each other? Three concepts were compared (soft locking, live merge with conflict banners, and a branch-and-merge model), and live merge with per-object presence and a personal undo stack was chosen. A low-fidelity wireframe set was refined into a high-fidelity Figma prototype. Testing with [N] users on [3] tasks gave a [X]% task success rate, and after one iteration, errors fell from [A] to [B] and satisfaction rose to [Y]/5.

1. Problem Definition and Scope

Problem statement

In real-time collaborative design tools, multiple users often edit the same object (a shape, text layer or component) at the same time. Current tools resolve this with "last write wins" and give little feedback, so users lose changes without knowing. This project designs how the tool shows presence, makes conflicts visible, resolves them, and defines what undo means when many people edit together.

Stakeholders and target users

- Primary: UI/UX designers and design students working in teams of 2–10.
- Secondary: developers and product managers who comment on or tweak designs.
- Stakeholders: team leads, clients reviewing files, tool vendors.

Goals and success metrics

- Users can identify who is editing an object within 3 seconds (target ≥90%).
- Users can recover a lost edit in under 30 seconds.
- Task success ≥85% for the 3 critical tasks.
- SUS score ≥75, and error rate down by ≥40% after iteration.

Constraints and assumptions

- Prototype only (no real sync engine); runs in Figma on desktop.
- Assumes stable internet normally, with offline handled as an edge state.
- Assumes users already know basic design tools.
- Time limit: one semester activity.

Scope: what the design will and will not cover

- Will cover: presence indicators, object-level edit awareness, conflict notification and resolution, undo/redo behaviour, version recovery.
- Will not cover: the backend sync algorithm (CRDT/OT), billing, full drawing tools, mobile app.

2. User Research

Research method and questions asked

Semi-structured interviews (15–20 min) and observation of a live two-person editing session.
Questions:

1. Describe the last time you edited a file with others at the same time.
2. Has anyone's change ever overwritten yours? What happened?
3. How do you know what a teammate is currently editing?
4. What do you expect Undo to do when others are also editing?
5. What do you do today to avoid conflicts (chat, duplicating files)?

Participant details

No.	Participant profile	Method	Key observation
P1	Isamoddin [CSE student, 20, uses Figma for projects]	Interview	[Lost a text edit with no warning]
P2	Deepak O K[Design student]	Interview + observation	[Pressed undo and removed teammate's change]
P3	Joseph[UI designer]	Interview	[Messages the team "I'm on this frame"]
P4	Chandan[Developer who edits specs]	Interview	[Unsure whose cursor is whose]
P5	Stalin[Project lead]	Interview	[Duplicates files to avoid overwriting]

Key findings and direct observations

- Users rely on external chat to claim objects, which shows the tool's presence cues are not enough.
- Silent overwrites cause the most frustration; users notice only later.
- Undo is feared: users don't know whether it affects only their actions.
- Colored cursors are seen but not linked to specific objects.

Competitor or existing solution analysis

Tool	Presence	Conflict handling	Undo	Gap
Figma	Cursors, avatars	Last write wins, no notice	Personal undo	No conflict message
Canva	Cursors, comments	Limited	Shared-feeling undo	Unclear ownership
Miro	Cursors, "locked" item	Soft lock option	Per-user	Locks block flow
Google Docs	Colored carets	Auto merge	Per-user	Not object-based

Insights that will drive the design

1. Awareness must be at object level, not just cursor level.
2. Conflicts must be visible at the moment they occur, with an easy recover path.
3. Undo must be personal, predictable and clearly labelled.
4. Resolving should not interrupt flow; blocking locks are disliked.

3. Personas and Empathy Map

Persona 1: Ananya, the Team Designer (22)

- Role: UI designer on a 5-person student team.
- Goals: finish screens fast, avoid rework.
- Behaviour: edits many frames at once, chats on WhatsApp to claim frames (from P2, P3).
- Frustrations: lost edits, unsure whose change is whose (P1, P2).
- Quote: "I only found out my text was gone when the lead asked."

Persona 2: Rohit, the Reviewer/Lead (26)

- Role: lead who tweaks spacing and text in others' frames.
- Goals: make quick fixes without breaking teammates' work.
- Behaviour: duplicates files to be safe (P5).
- Frustrations: afraid to use undo, cannot see history clearly (P2, P5).
- Quote: "I copy the file first. Otherwise I might wipe someone's work."

Empathy map: says, thinks, does, feels

- Says: "Who is editing this?", "Did you change the title?", "Don't touch frame 3."
- Thinks: "Will my edit survive?", "If I press undo, whose work disappears?"
- Does: messages teammates, duplicates files, saves versions manually.
- Feels: anxious, distrustful of the tool, annoyed at rework.

Pain points and unmet needs

- Silent overwrites → need conflict alerts
- Unclear ownership → need object-level presence
- Undo fear → need personal undo and recovery
- Communication outside the tool → need in-context signals

Problem reframed as a design question

How might we help collaborators see who is editing what, understand and resolve conflicts quickly, and undo with confidence, without slowing their flow?

4. User Journey and Information Architecture

End-to-end user journey(Ananya)

Stage	Action	Emotion
Open file	Joins shared file, sees teammates	😐 Neutral/positive
Pick object	Selects a card to edit	😊 Confident
Overlap	Teammate edits same card	😕 Confused (low)
Conflict	Her change is overwritten	😠 Frustrated (lowest)
Recovery	Searches version history	😟 Anxious
Resolve	Restores/merges change	😌 Relieved (high)

Task flows for the three most critical tasks

1. See who is editing an object: Open file → hover/select object → presence badge with name and edit state appears → user decides to wait or join.
2. Resolve a conflict: Both edit same property → conflict banner on the object → "Keep mine / Keep theirs / Keep both" → confirmation toast → change applied and logged.
3. Undo my last change safely: Press Undo → tooltip shows "Undo your change: Title text" → only own action reverted → teammate's edits stay → option to redo.

Sitemap or navigation structure

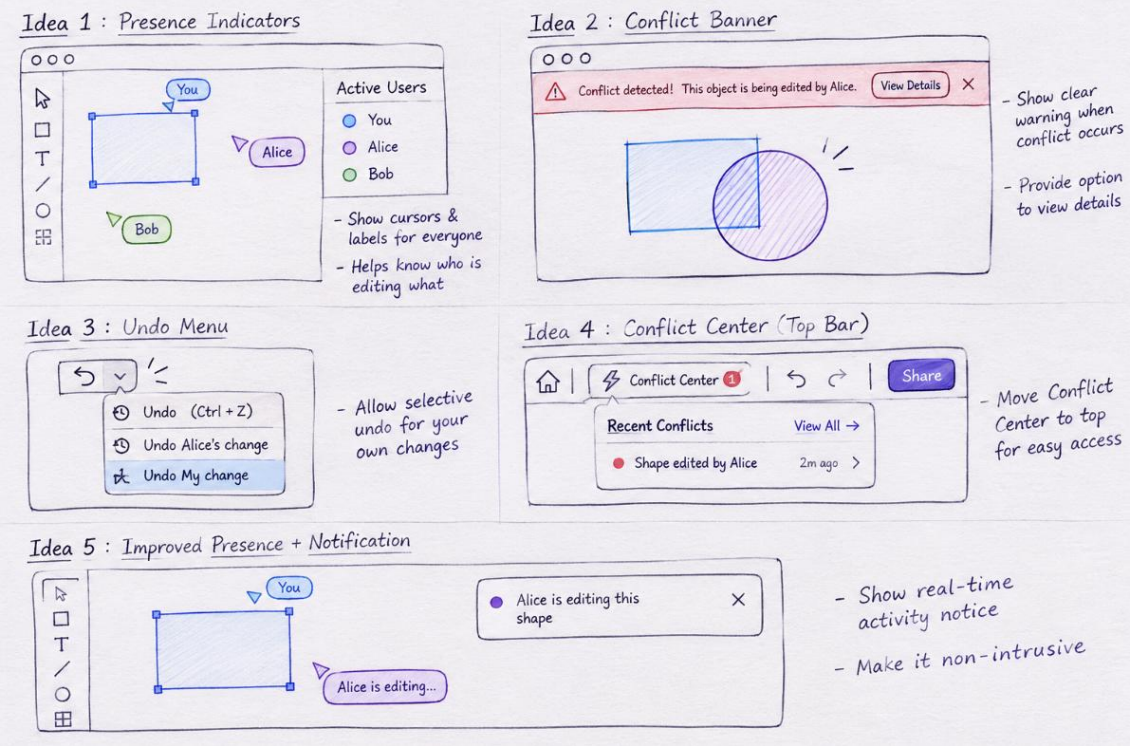
- Canvas
  - Presence bar (avatars, filters)
  - Object overlay (badge, edit state, conflict banner)
  - Right panel: Properties | Activity | Comments
- Activity & History (per-object timeline, restore)
- Conflict Center (list of unresolved conflicts)
- Settings (notifications, lock mode)

Content and feature prioritisation

Priority	Feature
Must	Object-level presence, conflict banner, personal undo, restore version
Should	Conflict Center, activity timeline, edit-intent hint
Could	Soft lock, @mention on conflict
Won't (now)	Branching, AI-assisted merge

5. Ideation and Concept Selection

Rough sketches or rapid ideas (include scanned pages)



Three distinct concepts

A: Soft Lock. Object shows "Ravi is editing"; others get a gentle warning and can override.

B: Live Merge with Conflict Banner. Everyone edits freely; conflicts show an inline banner with Keep mine / theirs / both. Personal undo stack.

C: Branch and Merge. Edits to contested objects go to a temporary branch that is merged via a review panel.

Evaluation of concepts against criteria

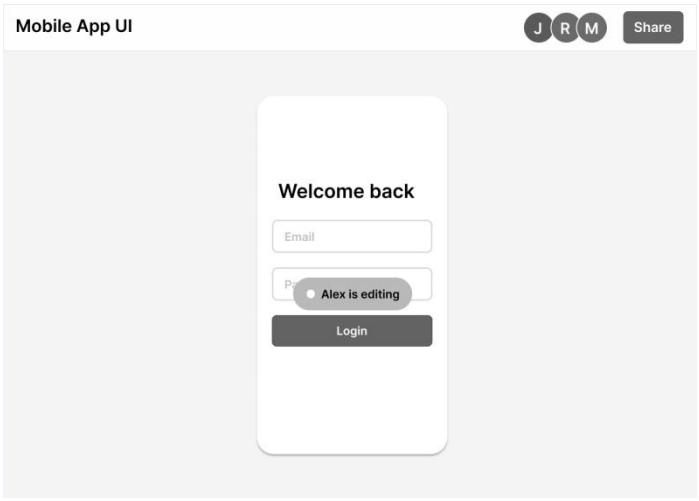
Criteria	Concept A	Concept B	Concept C
Preserves flow	2	5	2
Conflict visibility	3	5	4
Undo clarity	3	4	3
Learnability	5	4	2
Feasibility (prototype)	5	4	2
Total	**18**	**22**	**13**

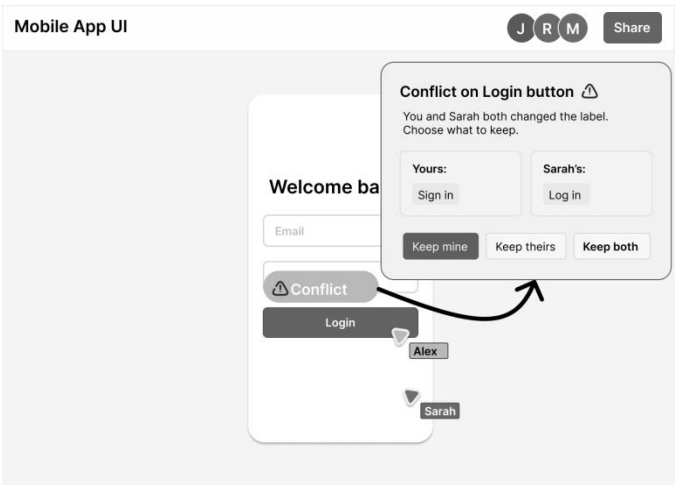
Chosen concept and reason for selection

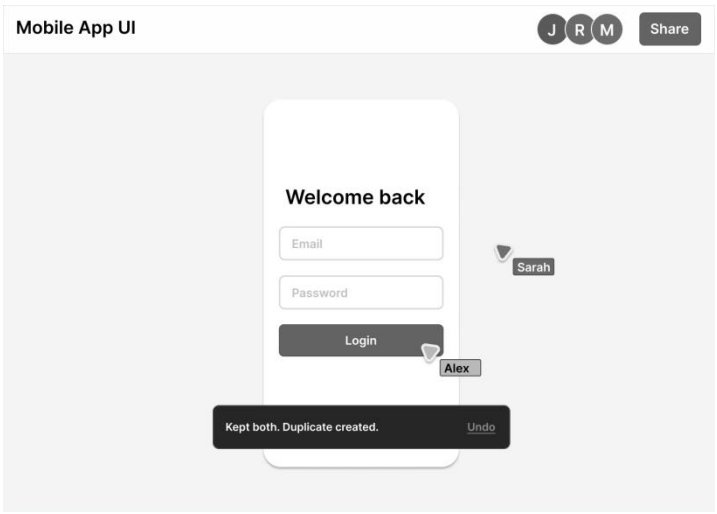
Concept B. It keeps users in flow (insight 4), makes conflicts visible at the moment they occur (insight 2), and supports personal undo (insight 3). Elements of A (a subtle "editing" badge) are borrowed for awareness.

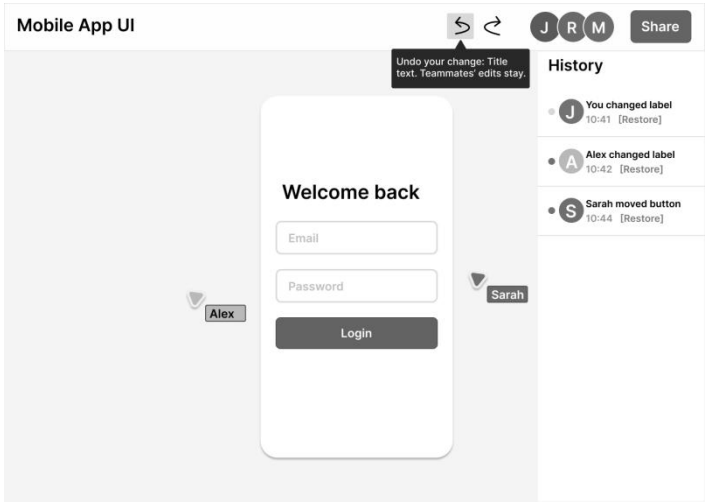
6. Low-Fidelity Wireframes

Wireframes of key screens









Annotations:

- Presence badge sits on the object's top-left corner so ownership is clear.
- The conflict banner is inline, not a modal, so work isn't interrupted.
- Keep mine / Keep theirs / Keep both have equal weight so no option is hidden.
- The undo tooltip names the exact action being undone.

Quick feedback from at least two people

- Reviewer 1 [Deepak O K/Student]: "The banner doesn't clearly explain why the conflict occurred. Adding more details would make it easier to understand."
- Reviewer 2 [Isam/Student]: "The Conflict Center should be easier to access, and the undo option should be more visible to users."

Changes made after feedback

- Added a "Why this happened" link in the banner.
- Moved Conflict Center to the top bar for visibility.
- Made the undo tooltip appear on hover as well as on press.

7. Visual Design and Design System

Colour palette with contrast ratios

Role	Hex	Use	Ratio on white
Primary	#2F5BFF	Actions, selection	~5.0:1
Success	#0B7A3E	Resolved	~5.4:1
Warning	#B45309	Conflict	~5.0:1
Error	#C62828	Failure	~5.6:1
Text	#1F2937	Body	~14:1
Surface	#F5F7FA	Panels	n/a

Typography and type scale

Inter. H1 28 / H2 22 / H3 18 / Body 14 / Caption 12; weights 400, 500, 600.

Spacing, grid and layout rules

8-pt spacing system (4, 8, 16, 24, 32); 12-column grid for panels; canvas fills the remaining space; minimum touch/click target 32 px.

Reusable components and their states

- Presence avatar: default, active, idle, offline
- Object badge: viewing, editing, conflict
- Conflict banner: default, hover, resolving, resolved
- Buttons: default, hover, focus, pressed, disabled
- Toast: success, info, error

Iconography and imagery choices

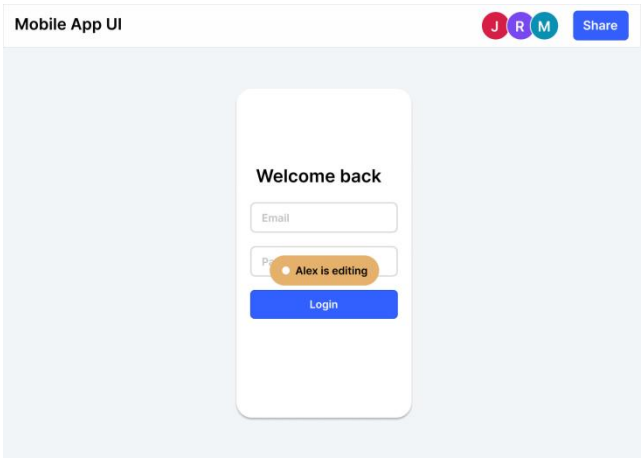
Outlined 1.5 px icon set (Lucide/Material); icons always have text labels; no decorative imagery to keep the canvas clean.

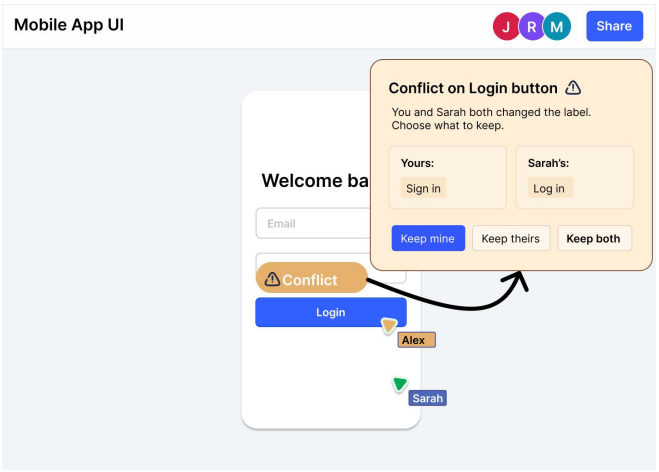
Rationale linking visual choices to user needs

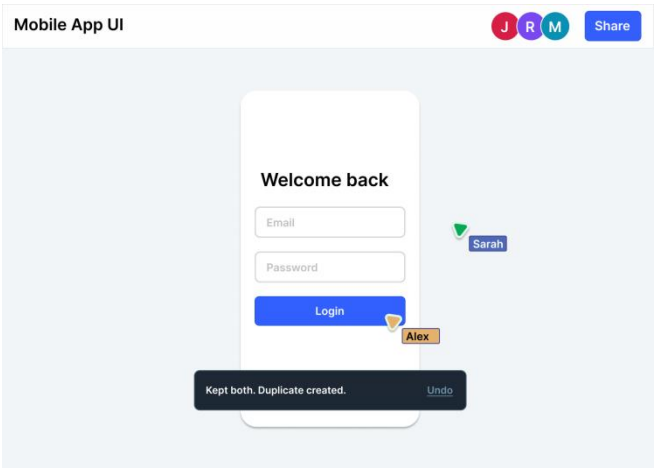
Warning amber with an icon and text makes conflicts noticeable without feeling like an error. Names plus colour support accessibility. A calm neutral UI keeps attention on the design canvas.

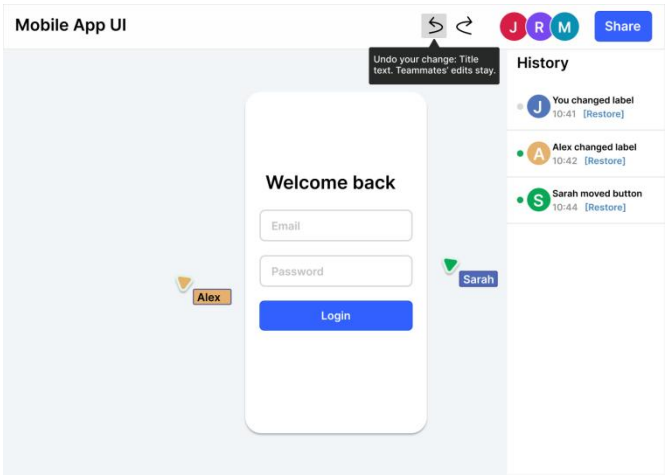
8. High-Fidelity Interactive Prototype

Final screens for all critical flows









Interactions, transitions and micro-interactions

- Badge fades in over 150 ms when a teammate selects an object.
- Banner slides in with a subtle pulse on the conflicting property.
- Toast confirms resolution with an "Undo" action for 5 seconds.
- Smart Animate used for state changes.

Edge states: empty, loading, error, offline and permission denied

- Empty: no collaborators → "You're editing alone. Invite teammates."
- Loading: skeleton for the history panel
- Error: "Couldn't sync your change. Retry" with local copy kept
- Offline: banner "Offline: changes will merge when you reconnect"
- Permission denied: view-only badge with "Request edit access"

Prototype link (Figma) and access instructions

Link:
https://www.figma.com/design/rtX50EoMwNtC6uKKcPxSY1/uiux_ConflictHandling_1EP24CS053?node-id=1-5&t=HebR9KNysynZugd4-1

Tools and resources used

Figma, FigJam (research synthesis), Google Forms (screening/SUS), WebAIM Contrast Checker, Stark/Figma A11y plugin, Lucide icons, Inter font.

9. Usability Testing and Iteration

Test plan

- Participants: 5+ users from the target group.
- Tasks: T1 identify who is editing the "Login button"; T2 resolve a text conflict and keep both versions; T3 undo your last change without affecting a teammate's.
- Success criteria: ≥85% completion, no more than 1 error per task, SUS ≥75.
- Method: moderated, think-aloud, 15 min each.

Results

User	Task	Success(Y/N)	Time	Issues observed
U1	T1	Y	12s	[none]
U2	T2	N	48s	[missed "Keep both"]
U3	T3	Y	20s	[hesitated at tooltip]

Accessibility check (WCAG 2.2 AA) and issues fixed

- Contrast ≥4.5:1 text, ≥3:1 UI components: verified
- Not colour-only: names/icons added to collaborator colours
- Keyboard: all banner actions reachable and focus visible
- Target size ≥24×24 px (2.5.8)
- Banner/toast announced via ARIA live regions (4.1.3)
- Issues fixed: [list what you actually found]

Problems found, ranked by severity

Severity	Problem
High	[Users missed "Keep both" option]
Medium	[Presence badge hidden on small objects]
Low	[Tooltip text too small]

Design changes made and evidence of improvement

[Change 1] → error rate from [A] to [B] in retest; [Change 2] → time on T2 from [x] s to [y] s; SUS from [x] to [y].

10. Design Rationale, Reflection and Future Work

Major design decisions and trade-offs

- Live merge over locking: preserves flow, but conflicts can still occur and need resolving.
- Inline banner over modal: less interruption, but may be missed, so a pulse animation was added.
- Personal undo over global undo: predictable for the user, but shared changes need separate "restore" via history.

Limitations of the final design

Tested only on desktop and with small groups; sync logic is simulated; complex multi-property conflicts are simplified; scalability beyond ~10 users is untested.

What failed or was redone, and what was learned

Concept A's lock was dropped after feedback that it blocked work. The first banner was a modal and was redone inline. Lesson: awareness works better than prevention, and you must test early with real users.

Ethical considerations

Presence tracking can feel like surveillance, so show activity only on the file, let users go "invisible" when idle, and avoid productivity scoring. Activity logs must be transparent. Participants gave informed consent and data was anonymised.

Future improvements

Mobile/tablet support, AI-suggested merges, property-level conflict detail, branch-and-merge for large changes, voice/comment threads on conflicts, accessibility for screen readers on canvas objects.

References (IEEE)

[1] Figma, “Multiplayer editing,” Figma Help Center. [Online]. Available: https://help.figma.com

[2] W3C, “Web Content Accessibility Guidelines (WCAG) 2.2,” 2023. [Online]. Available: https://www.w3.org/TR/WCAG22/

[3] J. Nielsen, “10 Usability Heuristics for User Interface Design,” Nielsen Norman Group, 1994. [Online]. Available: https://www.nngroup.com

[4] J. Brooke, “SUS: A quick and dirty usability scale,” in Usability Evaluation in Industry, London, U.K.: Taylor & Francis, 1996, pp. 189–194.

[5] D. Norman, The Design of Everyday Things, Rev. ed. New York, NY, USA: Basic Books, 2013.

[6] C. Sun et al., “Achieving convergence, causality preservation, and intention preservation in real-time cooperative editing systems,” ACM Trans. Comput.-Hum. Interact., vol. 5, no. 1, pp. 63–108, 1998.

Appendix

- Interview scripts and consent notes
- Raw research notes and survey responses
- Usability test scripts
- Additional screens and sketches